1.2 Sampling from a Population

- Outline
  1. Parameter versus Statistic
  2. Statistical Inference
  3. Sampling Bias
  4. Sampling Methods
     - Convenience Samples
     - Voluntary Response Samples
     - Simple Random Samples
     - Stratified Random Samples
  5. Other Sources of Bias

Terminology 4.1: Parameters

- A **parameter** is a numerical characteristic (e.g., the average value, the standard deviation, the largest, the smallest, etc.) of an entire population.
- Notes:
  1. When referring to a parameter, we usually use Greek letters. For instance, μ denotes the mean of a population, σ the standard deviation of a population, etc.
  2. Very seldom is the entire population available for observation.
     - Population might be *inaccessible*.
     - *Cost* of taking a complete census of the population might be prohibitive.
     - Decision must be made in a *timely* fashion.
  3. In practice, the *true* value of a parameter is most often unknown.

Terminology 4.2: Statistics

- A **statistic** is a numerical value summarizing the sample data.
- Notes:
  1. When referring to a statistic, we usually use alphabet letters. For instance, $\bar{x}$ denotes the mean of a sample, s the standard deviation of a sample, etc.
  2. We often use a statistic to estimate an unknown parameter. For example,
     - the sample mean $\bar{x}$ estimates the unknown value of a population mean μ.
     - the sample standard deviation s estimates the unknown value of a population standard deviation σ, etc.
  3. However, do **NOT** say $\bar{x} = \mu$ **NOR** $s = \sigma$.
  4. The value of a statistic changes from sample to sample. This sampling variability will be discussed in detail in Chapter 3.
  5. Use of a statistic in inferential problems will be discussed in Chapters 3–6.

Example 1.3: Parameters and Statistics

- To study the growth of pine trees at an early stage, a researcher records the heights (in cm) of 40 one-year-old red pine seedlings. The set of the 40 measurements averages 1.72.
    1. Is the number 1.72 a parameter or a statistic?
    2. The corresponding parameter is the mean height of *all* one-year-old pine seedlings.
    3. 1.72 estimates the population mean.
- A government agency wishes to assess the prevailing rate of unemployment in a county by sampling a small fraction of the labor force in the county and counting the number of persons currently unemployed. Suppose 500 randomly selected persons are interviewed and 40 are found to be unemployed. The proportion of the sample who are unemployed is $\frac{40}{500} = 0.08$.
    1. Is the number 0.08 a parameter or a statistic?
    2. The corresponding parameter is the proportion of unemployment in the *entire* county.
    3. The true value of the parameter is *unknown*.
    4. 0.08 estimates the population proportion, but it is subject to some *error* because it draws only from a part of the population.

A Big Picture of Statistical Inference

- Draw an image that illustrates how we do statistical inference on a population based on a sample.

“Good” Samples and Sampling Bias

- Our statistical inference on a population depends on a sample selected from the population.
- Selecting a “good” sample is critical to obtain accurate inference results on population.
- What is a good sample?
    1. Getting a good sample means choosing a good representative of the population.
    2. To get a good representative, a good sampling technique is necessary.
- A **biased sampling method** often produces a statistic that systematically differs from the corresponding population parameter.
- If sampling bias exists, we cannot trust generalizations from the sample to the population.

Convenience Samples

- **Convenience sampling**: Just ask whoever is around.
- Example: "Man on the street" survey (cheap, convenient, often quite opinionated, or emotional $\rightarrow$ now very popular with TV "journalism").
- Which men, and on which street?
  1. Ask about gun control or legalizing marijuana "on the street" in Berkeley, CA or in a small town in Arizona, and you would probably get totally different answers.
  2. Even within an area, answers would probably differ if you did the survey outside a high school or a country western bar.
- **Bias**: Opinions limited to individuals present.

Voluntary Response Samples

- **Voluntary response sampling**: The respondents choose themselves, which tend to cause large bias, because people with strong (especially negative) opinions are most likely to respond.
  1. Often called “public opinion polls.”
  2. These are not considered valid or scientific.
- **Bias**: Sample design systematically favors a particular outcome.
- Example. A hospital emails all nurses asking them to complete an online survey about their experience with the new electronic health record (EHR) system.
  1. This is a voluntary response sample, because nurses choose whether or not to participate.
  2. **Bias**: Only the nurses who feel strongly (perhaps those very frustrated with the system or those who find it very helpful) are most likely to respond, while others may ignore the survey.

Probability Samples

- Particular units are chosen by chance.
- Notes:
  1. No one group should be over-represented or under-represented.
  2. Tend to give an approximate representative of the population across a large spectrum of characteristics.
  3. Failure to use probability sampling often results in bias, or systematic errors in the way the sample represents the population.
- Examples:
  1. Simple random sample
  2. Stratified random sample
  3. Cluster sample
  4. Systematic sample
  5. Multi-stage sample

Simple Random Samples

- A **simple random sample (SRS)** is made of randomly selected individuals. Each individual in the population has the same probability of being in the sample.
- How to get a simple random sample of size n:
  1. Construct a list of units in the population by assigning a number from 1 to N, the total number of units in the population, to each unit in the population.
  2. Randomly select n integers from $\{1, 2, \ldots, N\}$ by using a random digits table or by using a random number generating computer program.

Example 1.4: Simple Random Samples

- A firm wants to understand the attitudes of its minority managers toward its system for assessing management performance. Below is a list of all the firm's managers who are members of minority groups. Select a simple random sample of six persons to be interviewed about the performance appraisal system.

Agarwal	Cross	Gonzalez	Lujan	Richards
Alfonseca	Dewald	Gomez	Mourning	Rodriguez
Baxter	Fleming	Hernandez	Nunez	Santiago
Bowman	Fonseca	Huang	Peters	Shen
Brown	Gates	Kim	Pliego	Vega
Cortez	Goel	Liu	Puri	Watanabe

  What is your simple random sample?

Stratified Random Samples

- A **stratified random sample** is a collection of SRSs performed on subgroups of a population.
- The subgroups are chosen to contain all the individuals with a homogeneous characteristic.
  1. Divide the population of NAU students into males and females.
  2. Divide the population of Arizona by major ethnic group.
  3. Divide the counties in America as either urban or rural based on criteria of population density.
- How to get a stratified random sample:
  1. Divide the population into homogeneous subgroups (**strata**) on some characteristic known and/or suspected to affect the response variables.
  2. Select a SRS from each of the strata.
  3. The collection of these selected SRSs constitutes a stratified random sample.
- In a stratified random sample, the SRS taken from each group need not be of the same size.
  1. A stratified random sample of 100 male and 150 female NAU students
  2. A stratified random sample of a total of 100 Arizonans, representing proportionately the major ethnic groups

Example 1.5: Stratified Random Samples

- A population of election districts might be divided into urban, suburban, and rural strata. Then select three simple random samples from each of these strata. Gathering these three simple random samples makes a stratified random sample.

Exercise 1.3

- A marketing research firm wishes to determine if the adult men in Flagstaff would be interested in a new upscale men's clothing store. From a list of all residential addresses in Flagstaff, the firm selects a simple random sample of 100 and mails a brief questionnaire to each. The population of interest is

  1. all adult men in Flagstaff.
  2. all residential addresses in Flagstaff.
  3. the members of the marketing firm that actually conducted the survey.
  4. the 100 addresses to which the survey was mailed.

  The sample in this survey is

  1. all adult men in Flagstaff.
  2. all residential addresses in Flagstaff.
  3. the members of the marketing firm that actually conducted the survey.
  4. the 100 addresses to which the survey was mailed.

Exercise 1.3 (continued)

- A television station is interested in predicting whether or not voters in its listening area are in favor of federal funding for abortions. It asks its viewers to phone in and indicate whether they support or are in favor of or opposed to this. Of the 2,241 viewers who phoned in, 1,574 (70.24%) were opposed to federal funding for abortions. The sample obtained is
  1. a simple random sample.
  2. a stratified random sample.
  3. a probability sample in which each person in the population has the same chance of being in the sample.
  4. probably biased.
- To take a sample of 90 members of a local gym, I first divide the members into men and women, and then take a simple random sample of 45 men and a separate simple random sample of 45 women. This is an example of
  1. a block design.
  2. a double-blind study.
  3. a simple random sample.
  4. a stratified random sample.

Realities of Sampling

- While a random sample is ideal, often it isn't feasible. A list of the entire population may not be available, or it may be impossible or too difficult to contact all members of the population.
- Sometimes, your population of interest has to be altered to something more feasible to sample from. Generalization of results are limited to the population that was actually sampled from.
- In practice, think hard about potential sources of sampling bias, and try your best to avoid them.

Other Forms of Bias

- Even with a random sample, data can still be biased, especially when collected on humans.
- Other forms of bias to watch out for in data collection:
  1. Question wording
  2. Context
  3. Inaccurate responses
  4. Many other possibilities. Examine the specifics of each study!

Question Wording (Allow vs. Not Forbid)

- "Do you think the US should allow public speeches against democracy?"

  $\boxed{\text{21\% said speeches should be allowed}}$

- "Do you think the US should not forbid public speeches against democracy?"

  $\boxed{\text{39\% said speeches should be not be forbidden}}$

Source: Rugg, D. (1941). Experiments in wording questions, Public Opinion Quarterly, 5, 91–92

Question Wording (The Impact of Details)

- A random sample was asked: “Should there be a tax cut, or should money be used to fund new government programs?”

  $\boxed{\text{Tax cut: 60\% vs. Programs: 40\%}}$

- A different random sample was asked: “Should there be a tax cut, or should money be spent on programs for education, the environment, health care, crime-fighting, and military defense?”

  $\boxed{\text{Tax cut: 22\% vs. Programs: 78\%}}$

Context

- Ann Landers column asked readers: “If you had it to do over again, would you have children?”
  1. The request for data contained a letter from a young couple which listed worries about parenting and various reasons not to have kids

     30% said “yes”

  2. Newsday conducted a random sample of all US adults, and asked them the same question, without any additional leading material

     91% said “yes”

Inaccurate Responses

- Do you think the following numbers are accurate?
  1. In a study on US students, 93% of the sample said they were in the top half of the sample regarding driving skill.[1]
  2. From a random sample of all US college students, 22.7% reported using illicit drugs.[2]

[1] Source: Svenson, O. (February 1981). Are we all less risky and more skillful than our fellow drivers?, Acta Psychologica 47(2), 143–148

[2] Source: Substance Abuse and Mental Health Services Administration (2010). Results from the 2009 National Survey on Drug Use and Health: Volume 1. Summary of National Findings (Office of Applied Studies, NSDUH Series H-38A, HHS Publication No. SMA 10-4856Findings). Rockville, MD, heeps://nsduhweb.rti.org/

Remarks

- **Always think critically about how the data were collected, and recognize that not all forms of data collection lead to valid inferences.**
- This is the easiest way to instantly become a more statistically literate individual!